Code Test 1

(1) For each layer, here is how we calculate ~~the~~ number of MACs :-

- For the ~~first~~ layer (784) :-

As there is no input we can't compute the number of MACs.

- For the ~~second~~ layer (256) :-

The input here is 1st layer (784)

The formula :- $x \times n$, [where x is number of inputs ~~and w is weight~~ and n is number of nodes]

$\Rightarrow$ $x = 784$ and $n = 256$

As per formula $= x \times n$

$= 784 \times 256$

$= \underline{\underline{200,704}}$

No. of MACs for layer (256) is $\underline{200,704}$.

• For the ~~third~~ layer (128):-

Here, $x = 256$ and $n = 128$

No. of MACs for Layer (128) $= 256 \times 128$
$= \underline{\underline{32,768}}$

• For the layer (10):-

Here, $x = 128$ and $n = 10$

No. of MACs for Layer (10) $= 128 \times 10$
$= \underline{\underline{1280}}$

(2) Sum of MACs across all three layer,

~~200704~~

$= 200704 + 32768 + 1280$
$= \underline{\underline{234752}}$

(3) Since, the network has no bias, the trainable parameters consists exclusively of the connection weights between layers.

Layer 1: 784 (inputs) × 256 (neurons) = 200704

Layer 2: 256 (inputs) × 128 (neurons) = 32768

Layer 3: 128 (inputs) × 10 (neurons) = 1280

Total parameters => 200704 + 32768 + 1280
= 234752 weights

(4) Since each weight is ~~sp~~ stored in FP32 format, it requires 4 bytes of memory.

Total weights = 234752

Total weight memory = 234752 × 4
= 939008 bytes

(5) Total Activation Memory requires storing the inputs and all layer outputs simultaneously. Each node is in FP32 format (4 bytes).

Total nodes => 784 + 256 + 128 + 10
= 1178 nodes

Total Activation Memory = 1178 × 4
= 4712 bytes

(6) Arithmetic Intensity measures how much math is done for every byte of data moved, using the provided formula:-

$$\frac{(2 \times \text{total MACs})}{\text{weight bytes} + \text{activation bytes}}$$

Calcuta

$\Rightarrow$ Arithmetic Intensity $= \dfrac{2 \times 234752}{939008 + 4712}$

$$= \frac{469504}{943720}$$

$$= 0.497$$